

Anil Seth



## 'Internet in a Box' or Craving for a Better Performance

This article walks the reader through the 'Internet in a box' and suggests how ISPs can improve download speeds. It also touches upon the Remy project, which generates algorithms to optimise the Internet's performance.

I came across the site <http://internet-in-a-box.org/>, which promises to create a small device with a local copy of a terabyte of the Internet's free information. The purpose of this project is to make a wealth of Internet resources available in environments that do not have a connection to the Internet, particularly, schools.

I was reminded of this project as I waited patiently for a streaming video to buffer. This was even after getting a higher bandwidth Net connection. You would not expect the 'Internet in a box' to resolve this performance problem if you installed this device. But what if an ISP used such a device? Would your experience of streaming video be better?

Would the experience of Stockholm's efforts at traffic decongestion be relevant at all? Stockholm introduced a charge for vehicles entering the central city and improved the commuting time. The results are explained in a very nice talk by Jonas Eliasson on [ted.com](http://ted.com) as follows:

*"Which means that we are now in a position where we have reduced traffic across this toll cordon with 20 per cent, and reduced congestion by enormous numbers, and people aren't even aware that they have changed, and they honestly believe that they have liked this all along."*

A relevant lesson is that congestion is a non-linear phenomenon and a small decrease in traffic can have a substantial impact on congestion.

### Open source downloads

Your current experience may have been that when updating a Linux distribution, the speeds are better if the servers are in the US! There are hardly any mirrors in India. The shortest or fastest data path to a mirror in the neighbouring countries might well be via the US.

An ISP can legally and morally host or mirror such open source content. All the open source distributions encourage creation of new mirrors. An ISP can deliver all requests for such content from its local copy. Unfortunately, an ISP may feel that the volume of traffic is not high enough to make hosting a mirror worth the expense and effort.

But, the experience with Stockholm's traffic could be a guideline. The realisation that the congestion is non-linear may imply a far greater impact on latency than the quantum of open source download traffic might indicate.

If local mirroring helps, then all the ISP customers benefit. Each one will experience fewer instances of degraded performance, especially when dealing with streaming content. Of course, as a reader of this magazine, you will benefit more because you are likely to be a consumer of open source content.

The ISP will gain as well. The first gain is that an ISP's external bandwidth requirement may be lower for the same level of performance. The second gain is that its revenues will increase, if my personal experience is anything to go by. I, in fact, reduced my bandwidth subscription because of latency and couldn't consume the bandwidth I had wanted to use.

You may also have ignored viewing a lot of content because it is irritating to watch it in a start and stop mode. Besides, time is a precious commodity. If you spend more time to watch one video, you have less time for additional content. An obvious possibility is that if you could view more, you would be willing to subscribe to a higher bandwidth and pay more to the ISP.

It would have been great to simulate the overall impact of downloading from a local mirror in contrast to downloading from an external mirror. Unfortunately, this simulation needs the data and the resources of an ISP. It would be nice if there was a way to nudge an ISP to think along these lines.

An alternate hope for better Internet performance is Remy, a project by MIT, which generates TCP congestion-control algorithms that "significantly outperformed algorithms devised by human engineers." ('A faster Internet—designed by computers?' - MIT News)

There is hope. It wasn't long ago that you may have disabled images to get reasonable browsing experience. Today, you are able to view a fair number of videos. If the past is anything to go by, you can be certain that the performance next year will be better than it is today. The wish, though, is that why can't the improvements be visible tomorrow! 

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